

Which vehicles to buy, and why

A plain-language answer to the question:

“How do I spend \$1.55 million on a delivery fleet and get the most capability out of it?”

THE SHORT VERSION

We can buy you **63 vehicles for \$1,548,641** with a combined **7,146 horsepower** that meets your **30 mpg** sustainability target — but only if you allow eleven of them to fall **four horsepower** below your towing threshold.

Hold that threshold firm and the 30 mpg target becomes impossible. Not expensive — *impossible*. No budget fixes it.

01 What you asked us for

You gave us six rules. Here they are as we understood them, in your own words, alongside what we did with each one.

WHAT YOU SAID	HOW WE HANDLED IT
“I’ve got about \$1.55M”	Treated as a hard ceiling. We never propose spending a dollar more.
“As much total engine power as possible”	This became the goal. Everything else is a boundary; this is the thing we push as high as it will go.
“Every vehicle needs at least 100 hp for towing”	Used as a filter. Vehicles under 100 hp were removed from consideration entirely — which turned out to be the pivotal decision in this whole report.
“At least 60 vehicles”	Hard floor on fleet size.
“No more than a dozen of any one model”	Cap of 12 units per model.
“Corporate wants the fleet to average at least 30 mpg”	Treated as a <i>target</i> rather than an absolute rule, because you said “wants”. That reading matters — see section 3.
“Prefer European manufacturers if you can”	Noted as a preference, not a rule. We could not enforce it strictly (see section 5), but our recommendation happens to include twelve European vehicles.

What we searched

The vehicle catalogue contains 398 listings covering 301 distinct models. After removing listings with missing engine data and everything below your 100 hp towing threshold, **132 models** remained eligible. We then examined every possible combination of those 132 models — every mix, every quantity — and identified the single best one. This is not a sample or an estimate. It is the provably best answer available from this catalogue under your rules.

02 Option A — your rules, exactly as written

If we hold every one of your rules firm, this is the best fleet money can buy. It is a good fleet. It misses your fuel-economy target by a hair.

60	6,804	\$1.54M	29.8			
VEHICLES	TOTAL HORSEPOWER	TOTAL SPEND	AVG MPG (TARGET 30)			
MODEL	UNITS	HP EACH	MPG	PRICE EACH	SUBTOTAL	ORIGIN
Datsun 280-ZX	12	132	32.7	\$27,354	\$328,248	JAPAN
Chevrolet Citation	12	115	28.8	\$24,393	\$292,716	USA
Oldsmobile Omega Brougham	12	115	26.8	\$25,380	\$304,560	USA
Dodge Colt	12	105	27.9	\$26,320	\$315,840	USA
Datsun 200SX	12	100	32.9	\$24,581	\$294,972	JAPAN
Total	60	6,804	29.82		\$1,536,336	

This leaves **\$13,664 unspent**. That is not an oversight and it is not a rounding error — there is genuinely nothing left in the catalogue worth buying with it. Any additional vehicle you could afford would drag the fleet average below 29.8 mpg, and swapping in anything more powerful would push you over budget. The money has nowhere useful to go.

NOTICE WHAT IS MISSING

There are no European vehicles in this fleet. Under your rules as literally written, none of them earn a place. Section 4 shows how a very small concession changes that.

03 The one rule we could not meet

Your 30 mpg fleet average cannot be achieved. We want to be precise about why, because the reason is not the one people usually assume.

Once you require every vehicle to produce at least 100 horsepower, the most fuel-efficient vehicle left in the entire catalogue manages **32.9 mpg**. There are exactly a handful of models near that mark, and you

are capped at twelve units of each. Build the most economical fleet those constraints permit and you land at **29.82 mpg**.

You are short by 0.18 mpg. Less than one percent. And there is nothing you can do about it.

MORE MONEY DOES NOT HELP

This is the finding we most want you to take away. We re-ran the analysis at progressively larger budgets to see where the target becomes reachable:

IF YOUR BUDGET WERE...		CAN YOU HIT 30 MPG WITH A 100 HP FLOOR?
\$1,550,000	YOUR BUDGET	No
\$1,860,000	+20%	No
\$2,500,000	+61%	No
\$5,000,000	+223%	No

Money is not your constraint. You are not short of budget — you are short of *catalogue*. Vehicles that tow well and sip fuel simply do not exist in the range available to you, and no amount of spending conjures them into existence. Going back to Finance for more money would waste everyone's time.

What fuel economy actually costs you

Since the target is negotiable in principle, here is the exchange rate. Each additional mile per gallon you demand of the fleet costs you roughly 450 to 550 horsepower, because you must substitute economical vehicles in place of powerful ones.

IF THE FLEET TARGET WERE...	TOTAL HORSEPOWER	POWER GIVEN UP	SPEND
27.0 mpg	8,245	—	\$1,549,787
28.0 mpg	7,805	−440	\$1,545,821
29.0 mpg	7,347	−458	\$1,547,033
29.8 mpg THE CEILING	6,804	−543	\$1,536,336

If corporate will accept 28 mpg instead of 30, you gain roughly a thousand horsepower for the same money. That is a conversation worth having before you sign anything.

04 Option B — our recommendation

We tested what happens if the towing threshold moves. The result is unusual enough that it is the main reason this report exists.

Lower the towing floor from 100 hp to 95 hp — a four-horsepower concession affecting eleven vehicles — and three things happen at once. You meet the 30 mpg target. You gain 342 horsepower. And twelve European vehicles enter the fleet.

63	7,146	\$1.55M	30.0			
VEHICLES (+3)	HORSEPOWER (+5%)	TOTAL SPEND	AVG MPG (TARGET MET)			
MODEL	UNITS	HP EACH	MPG	PRICE EACH	SUBTOTAL	ORIGIN
Datsun 280-ZX	12	132	32.7	\$27,354	\$328,248	JAPAN
Chevrolet Citation	12	115	28.8	\$24,393	\$292,716	USA
BMW 2002	12	113	26.0	\$21,000	\$252,000	EUROPE
Datsun 200SX	12	100	32.9	\$24,581	\$294,972	JAPAN
Oldsmobile Omega Brougham	3	115	26.8	\$25,380	\$76,140	USA
Buick Estate Wagon	1	225	14.0	\$29,008	\$29,008	USA
Toyota Celica GT	96 HP — BELOW FLOOR	11	96	32.0	\$25,051	JAPAN
Total	63	7,146	30.02		\$1,548,641	

Why relaxing a rule makes the fleet stronger

This is counter-intuitive, so it is worth a moment. The 30 mpg target applies to the fleet *average*, not to each vehicle. The eleven Toyotas are extremely economical at 32 mpg, which creates headroom in the average — and that headroom lets you buy genuinely powerful vehicles elsewhere, including a 225 hp Buick that your original rules could never have afforded to accommodate. Eleven slightly weaker vehicles buy you the freedom to make the other fifty-two considerably stronger.

	OPTION A	OPTION B	DIFFERENCE
Vehicles	60	63	+3
Total horsepower	6,804	7,146	+342
Fleet average mpg	29.82	30.02	target met
Total spend	\$1,536,336	\$1,548,641	+\$12,305
European vehicles	0	12	+12
Vehicles below 100 hp	0	11	+11

THE DECISION IN FRONT OF YOU

Everything hinges on one question: **is the 100 hp towing threshold a hard engineering requirement, or a round number somebody picked?**

If it is a real limit — a trailer weight, a legal classification, a safety certification — take Option A and tell corporate the 30 mpg target is unattainable at any budget. If it is a rule of thumb, take Option B. You get more capability, you satisfy sustainability, and you get the European vehicles you wanted, for twelve thousand dollars more.

05 What we assumed

Every number in this report rests on the assumptions below. If any of them is wrong, tell us and we will re-run the analysis in under a minute.

Prices are estimated, not quoted

This is the assumption most worth your attention. **The catalogue contains no prices.** It lists engine power, weight, fuel economy, model year and origin — but not what anything costs. Rather than abandon the analysis, we estimated each vehicle's price from its curb weight at **\$9.40 per pound**, which puts a typical vehicle in this catalogue at roughly \$26,000.

The recommendations hold as long as heavier vehicles genuinely cost more than lighter ones, which is broadly true. But the exact dollar figures are estimates. Give us a real price list and the answer may shift — the method will not.

Everything else

- “About \$1.55M” was read as a firm ceiling of exactly \$1,550,000. If you can flex to \$1.6M, say so; it does not change the fuel-economy finding, but it may change the mix.
- The 30 mpg target was applied to the *fleet average*, not to each vehicle individually. Applied per-vehicle it would be far more restrictive, and Option B would not exist. Please confirm which you meant.
- Where the same model appeared several times in the catalogue at different weights, we used the cheapest listing.
- Six listings had no engine-power figure recorded and were excluded rather than guessed at.
- Fuel economy figures are the catalogue's own, unadjusted for load, terrain or duty cycle. A working delivery fleet will do worse in practice — the *comparison* between options remains valid, but treat 30 mpg as a catalogue figure rather than a promise.

What we could not answer

You asked us to prefer European manufacturers. We could not enforce this properly. The catalogue records only a broad region — USA, Japan or Europe — with no manufacturer detail beneath it, and no way to distinguish, say, a German manufacturer from an Italian one. We therefore treated it as a tie-breaker rather than a rule. Option B includes twelve European vehicles as a consequence of the numbers, not because we steered toward them. If European sourcing matters to you commercially, we need a supplier list to do it justice.

How this analysis was produced. An AI agent read the raw vehicle catalogue, worked out for itself which columns represented cost, power and efficiency, translated your written instructions into a formal mathematical model, and handed that model to an optimisation engine that searched the full space of possible fleets. When your original rules turned out to be unsatisfiable, the agent identified precisely which rule was responsible and by how much it would need to bend, rather than silently discarding it. Every figure in this report was produced by that engine and can be reproduced exactly. Nothing here was estimated by eye.